Analyzing Thematic Journal Alignment

Alessandro Macchi (53633A)
Natural Language Processing
MSc in Data Science for Economics

Table of contents

1 **Introduction**

Project goal and research question

2 **Methodology**

How the pipeline works

3 **Experimental Results**

Alignment trends, outliers and topic patterns

4 **Conclusions**

Key insights and limitations

Introduction

Subject

PubMed – World Journal of Orthopedics

Research Que

To what extent do a
the World Journal
show semantic si
journal's Aims &

Metric

Cosine similarity between article abstract and the journal’s Aims & Scope

Objectiv

Build a reusabl
pipeline to analy
PubMed journal

Methodology

Preprocessing

Abstracts are collected, cleaned and filtered to create the final article corpus.

Text representation

The corpus is transformed into SentenceTransformer embeddings, using chunk averaging for long texts.

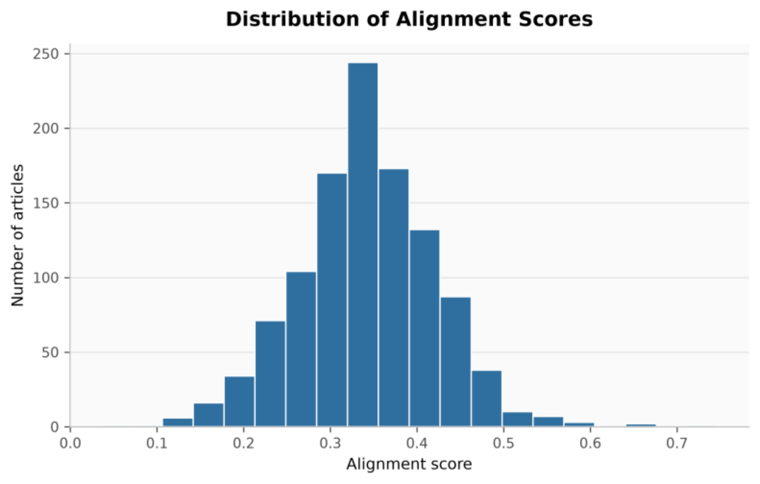
Alignment scoring

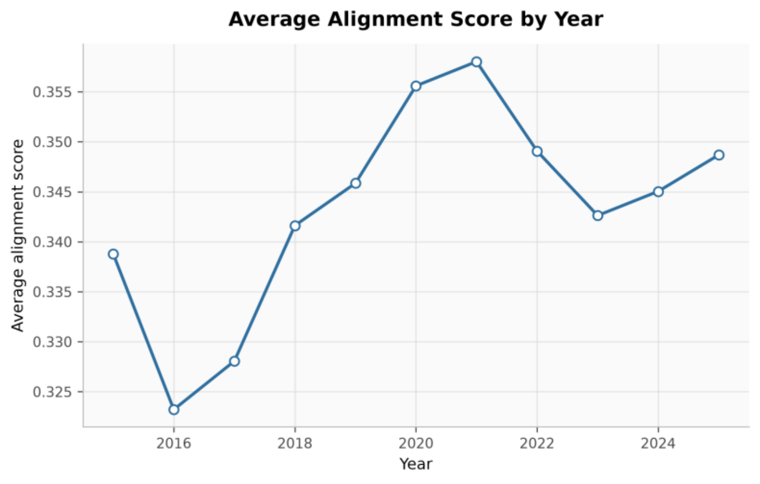
Cosine similarity is used to measure thematic alignment.

Analysis

Scores are aggregated over time, outliers are detected with z-scores with the usage of BERTopic

Score Results





Score distribution

Most articles are concentrated around analogous alignment values, suggesting that the journal's publications are similarly coherent with its Aims & Scope.

Score per year

The yearly trend shows a mild increase up to 2021, followed by a slight decrease in 2022–2023 and a stabilization around the central range in 2024–2025.

Outlier Analysis

The analysis identifies 47 outliers, which could be divided in two groups:

High-alignment

Usually, broad articles that discuss the journal itself or general orthopedic topics

Low-alignment

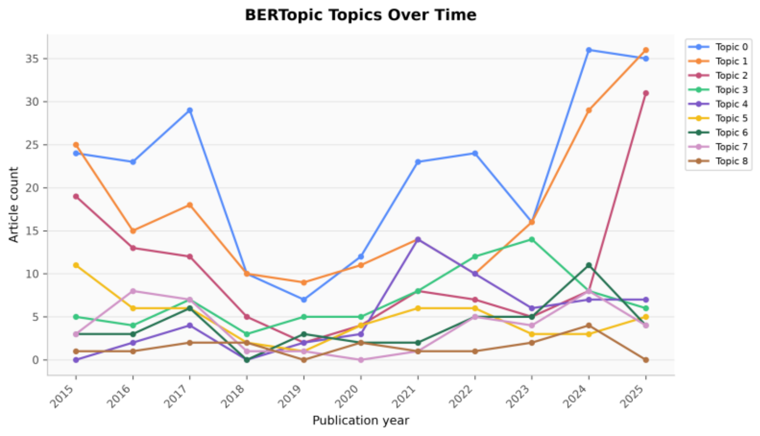
Articles still relevant to orthopedics, but the metric gives them lower scores because their terminology is less similar to the Aims & Scope.

BERTopic Results

Topic	Main keywords	Abstracts	Articles (%)	Mean Score
Noise	patients; bone; treatment; results; methods	267	24.27	0.331
G0	fractures; patients; hip; fracture; femoral	239	21.73	0.349
G1	knee; patients; tka; group; bone	193	17.55	0.363
G2	spinal; pain; lumbar; patients; spine	114	10.36	0.322
G3	infection; pji; antibiotic; patients; infections	77	7.00	0.313
G4	covid19; orthopaedic; pandemic; orthopedic; ai	55	5.00	0.403
G5	shoulder; cuff; rotator; tears; repair	53	4.82	0.329
G6	ankle; foot; patients; pain; tendon	44	4.00	0.380
G7	acl; reconstruction; meniscal; graft; ligament	42	3.82	0.309
G8	bone; treatment; lesions; patient; abc	16	1.45	0.310

The first three clusters are broader and closer to the overall mean score.
The remaining six clusters are more specific and allows for more interesting insights.

The decrease in 2022–2023 is partly explained by the growth of G3, which has a negative effect on the average score, and by the decline of G4, which has a positive effect.
G7 has the lowest alignment score, but its keywords are still clearly coherent with orthopedics.



Conclusion

Results

Temporal analysis reveals some insights, thanks to topic composition shifts captured by BERTopic.

Limitations

Cosine similarity is sensitive to lexical overlap with the Aims & Scope statement and does not have a direct interpretation.

Future work

Replacing all-MiniLM-L6-v2 with a domain-adapted model such as BioBERT or enriching the reference document with representative abstracts.

Thank you for the attention!